

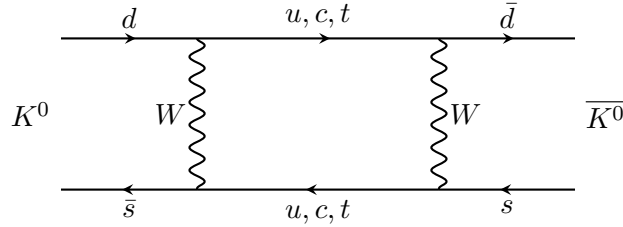
B4 Nuclear and Particle Physics — Model Solutions, Trinity 2022

Question 1 — Neutral kaon mixing and oscillation

Problem. (a) Leading-order $K^0 \leftrightarrow \bar{K}^0$ mixing diagram. (b) From $|\pm\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle \pm |\bar{K}^0\rangle)$ being CP -eigenstates, deduce C , the pion content of each decay, draw diagrams, and explain $\tau_S \ll \tau_L$. (c) Threshold for $pp \rightarrow p\pi^+\Lambda^0 K^0$, quark-flow diagram, regeneration. (d) Justify $A_S(t) = A_S(0)e^{-im_S t}e^{-t/(2\tau_S)}$ and write $A_{K^0}, A_{\bar{K}^0}$. (e) For initial pure K^0 , find A, B, C, ϕ and sketch.

(a)

$K^0 = (d\bar{s})$, $\bar{K}^0 = (\bar{d}s)$: $|\Delta S| = 2$ requires second-order weak box diagram with internal u, c, t .



(b)

CP eigenvalues by hypothesis:

$$CP|+\rangle = +|+\rangle, \quad CP|-\rangle = -|-\rangle.$$

Multiply the defining relations by $\sqrt{2}$:

$$|K^0\rangle + |\bar{K}^0\rangle = \sqrt{2}|+\rangle, \quad |K^0\rangle - |\bar{K}^0\rangle = \sqrt{2}|-\rangle.$$

Apply CP to both:

$$CP(|K^0\rangle + |\bar{K}^0\rangle) = +(|K^0\rangle + |\bar{K}^0\rangle).$$

$$CP(|K^0\rangle - |\bar{K}^0\rangle) = -(|K^0\rangle - |\bar{K}^0\rangle).$$

Add the two equations and divide by 2:

$$CP|K^0\rangle = |\bar{K}^0\rangle.$$

Subtract and divide by 2:

$$CP|\bar{K}^0\rangle = |K^0\rangle.$$

Use $P|K^0\rangle = -|K^0\rangle$ (pseudoscalar, $J^P = 0^-$):

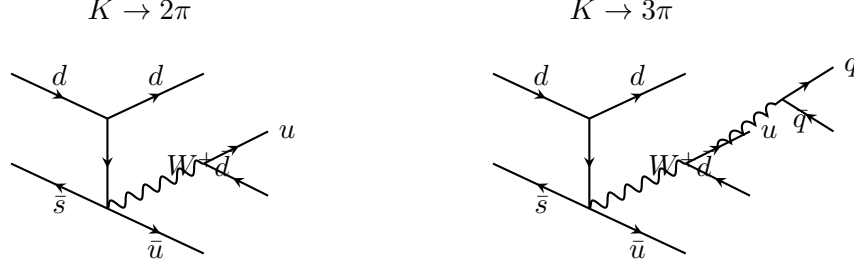
$$CP|K^0\rangle = C(-|K^0\rangle) = -C|K^0\rangle.$$

Combine with $CP|K^0\rangle = |\overline{K^0}\rangle$:

$$\boxed{C|K^0\rangle = -|\overline{K^0}\rangle, \quad C|\overline{K^0}\rangle = -|K^0\rangle.}$$

Pion content. For $J = 0$ kaons, $CP(n\pi) = (-1)^n$:

$$|+\rangle (CP = +1) \rightarrow 2\pi, \quad |-\rangle (CP = -1) \rightarrow 3\pi.$$



Lifetime ordering. Q -values:

$$Q_{2\pi} = m_K - 2m_\pi \approx 220 \text{ MeV}, \quad Q_{3\pi} = m_K - 3m_\pi \approx 80 \text{ MeV}.$$

Phase space drives Γ , so:

$$\boxed{K_S^0 \leftrightarrow |+\rangle \rightarrow 2\pi, \quad K_L^0 \leftrightarrow |-\rangle \rightarrow 3\pi, \quad \tau_L/\tau_S \sim 580.}$$

(c)

Threshold. Mandelstam s at threshold (all final-state particles at rest in c.m.):

$$\sqrt{s}_{\min} = m_p + m_{\pi^+} + m_\Lambda + m_{K^0} \equiv M.$$

Lab frame, target proton at rest, beam proton 4-momentum $(E_1, \vec{p}_1)$:

$$s = (E_1 + m_p)^2 - |\vec{p}_1|^2.$$

Expand the square:

$$s = E_1^2 + 2m_p E_1 + m_p^2 - |\vec{p}_1|^2.$$

Apply $E_1^2 - |\vec{p}_1|^2 = m_p^2$:

$$s = 2m_p E_1 + 2m_p^2.$$

Equate to M^2 at threshold:

$$2m_p E_{1,\min} + 2m_p^2 = M^2.$$

Solve for $E_{1,\min}$:

$$\boxed{E_{1,\min} = \frac{M^2 - 2m_p^2}{2m_p}, \quad T_{1,\min} = E_{1,\min} - m_p.}$$

Sum masses:

$$M = 938.3 + 139.6 + 1115.7 + 497.6 = 2691.2 \text{ MeV}/c^2.$$

Square:

$$M^2 = 7.242 \times 10^6 \text{ MeV}^2.$$

Compute $2m_p^2$:

$$2m_p^2 = 1.761 \times 10^6 \text{ MeV}^2.$$

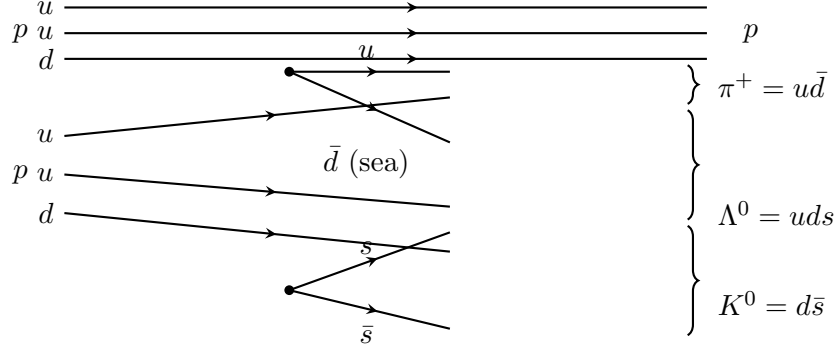
Plug in:

$$E_{1,\min} = \frac{(7.242 - 1.761) \times 10^6}{2 \times 938.3} \text{ MeV} = 2920 \text{ MeV}.$$

Subtract proton mass:

$$T_{1,\min} \approx 1.98 \text{ GeV}.$$

Quark flow. Associated production via sea $s\bar{s}$:



Regeneration. After many τ_S in vacuum:

$$|\psi\rangle = |K_L^0\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle).$$

Through matter, forward amplitudes differ ($\bar{K}^0 N \rightarrow \Lambda \pi$ is open, $K^0 N$ is not):

$$|\psi\rangle_{\text{out}} = \frac{1}{\sqrt{2}}(f|K^0\rangle - \bar{f}|\bar{K}^0\rangle).$$

Substitute $|K^0\rangle = (|K_S\rangle + |K_L\rangle)/\sqrt{2}$, $|\bar{K}^0\rangle = (|K_S\rangle - |K_L\rangle)/\sqrt{2}$:

$$|\psi\rangle_{\text{out}} = \frac{1}{2}(f(|K_S\rangle + |K_L\rangle) - \bar{f}(|K_S\rangle - |K_L\rangle)).$$

Collect K_S, K_L coefficients:

$$|\psi\rangle_{\text{out}} = \frac{f-\bar{f}}{2}|K_S^0\rangle + \frac{f+\bar{f}}{2}|K_L^0\rangle.$$

K_S component is regenerated (Pais–Piccioni).

(d)

Mass eigenstate: well-defined mass m_S and width $\Gamma_S = 1/\tau_S$. Stationary part: $e^{-im_S t}$. Decay of $|A|^2 \propto e^{-t/\tau_S}$ requires:

$$A_S(t) = A_S(0) e^{-im_S t} e^{-t/(2\tau_S)}, \quad A_L(t) = A_L(0) e^{-im_L t} e^{-t/(2\tau_L)}.$$

Equivalent to complex energy $E_{S,L} = m_{S,L} - i\Gamma_{S,L}/2$ (Wigner–Weisskopf).

In CP -conserving limit $K_S^0 = |+\rangle$, $K_L^0 = |-\rangle$:

$$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_S^0\rangle + |K_L^0\rangle), \quad |\bar{K}^0\rangle = \frac{1}{\sqrt{2}}(|K_S^0\rangle - |K_L^0\rangle).$$

Project:

$$A_{K^0}(t) = \frac{1}{\sqrt{2}}(A_S(t) + A_L(t)), \quad A_{\bar{K}^0}(t) = \frac{1}{\sqrt{2}}(A_S(t) - A_L(t)).$$

(e)

Project $|K^0\rangle = \frac{1}{\sqrt{2}}(|K_S^0\rangle + |K_L^0\rangle)$ to read off initial amplitudes:

$$A_S(0) = A_L(0) = 1/\sqrt{2}.$$

Substitute into part (d):

$$A_{K^0}(t) = \frac{1}{2}(e^{-im_S t} e^{-t/(2\tau_S)} + e^{-im_L t} e^{-t/(2\tau_L)}).$$

$$A_{\overline{K^0}}(t) = \frac{1}{2}(e^{-im_S t} e^{-t/(2\tau_S)} - e^{-im_L t} e^{-t/(2\tau_L)}).$$

Form $|A_{K^0}|^2 = A_{K^0} A_{K^0}^*$:

$$|A_{K^0}|^2 = \frac{1}{4}(e^{-im_S t} e^{-t/(2\tau_S)} + e^{-im_L t} e^{-t/(2\tau_L)})(e^{+im_S t} e^{-t/(2\tau_S)} + e^{+im_L t} e^{-t/(2\tau_L)}).$$

Expand the diagonal terms:

$$e^{\pm im_S t} e^{\mp im_S t} e^{-t/\tau_S} = e^{-t/\tau_S}, \quad e^{\pm im_L t} e^{\mp im_L t} e^{-t/\tau_L} = e^{-t/\tau_L}.$$

Collect the cross terms:

$$|A_{K^0}|^2 = \frac{1}{4}(e^{-t/\tau_S} + e^{-t/\tau_L} + e^{-t/(2\tau_S)} e^{-t/(2\tau_L)} [e^{i(m_L - m_S)t} + e^{-i(m_L - m_S)t}]).$$

Apply $2 \cos \theta = e^{i\theta} + e^{-i\theta}$:

$$|A_{K^0}|^2 = \frac{1}{4}(e^{-t/\tau_S} + e^{-t/\tau_L} + 2e^{-t/(2\tau_S)} e^{-t/(2\tau_L)} \cos[(m_L - m_S)t]).$$

Analogously for $\overline{K^0}$:

$$|A_{\overline{K^0}}|^2 = \frac{1}{4}(e^{-t/\tau_S} + e^{-t/\tau_L} - 2e^{-t/(2\tau_S)} e^{-t/(2\tau_L)} \cos[(m_L - m_S)t]).$$

Match to $I = Ae^{-t/\tau_S} + Be^{-t/\tau_L} + C \cos \phi$:

$$A = B = \frac{1}{4}, \quad C = \pm \frac{1}{2} e^{-t/(2\tau_S)} e^{-t/(2\tau_L)}, \quad \phi = \Delta m t,$$

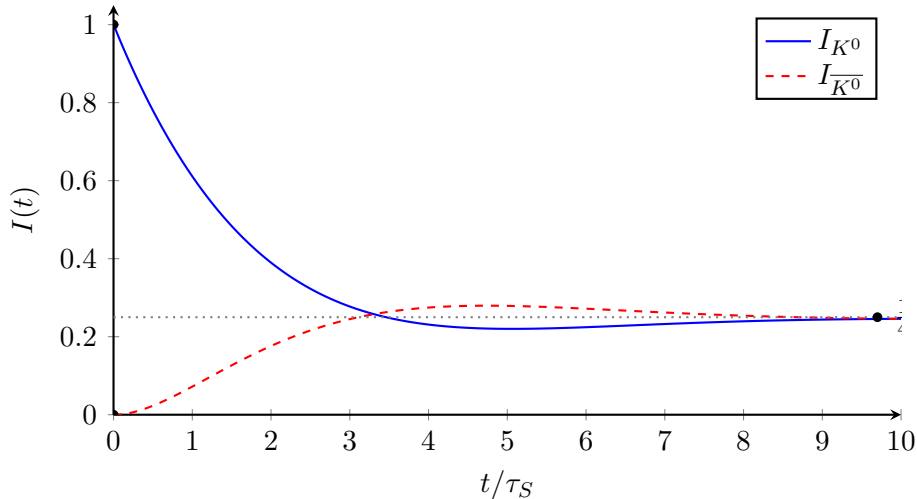
+ for K^0 , - for $\overline{K^0}$, $\Delta m = m_L - m_S = 3.49 \times 10^{-12} \text{ MeV}/c^2$.

Numerical scales for sketch. In $\tilde{t} = t/\tau_S$:

$$\tau_S/\tau_L = 8.958 \times 10^{-11}/5.18 \times 10^{-8} = 1.73 \times 10^{-3}.$$

$$\Delta m \tau_S/\hbar = (3.49 \times 10^{-12} \text{ MeV})(8.958 \times 10^{-11} \text{ s})/(6.582 \times 10^{-22} \text{ MeV s}).$$

$$\Delta m \tau_S/\hbar \approx 0.475 \text{ rad}.$$



Question 2 — Fermi's Golden Rule and Feynman diagrams

Problem. (a) Define terms in $\Gamma = (2\pi/\hbar)|M|^2\rho$. (b) Diagram and phase-space estimate for $\pi^+ \rightarrow \mu^+\nu_\mu\gamma$. (c) Estimate τ_τ from τ_μ . (d) Relate diagrams to FGR. (e) Diagrams for Compton, bremsstrahlung, pair production. (f) Diagrams and cross-section ordering for four $e\nu$ channels. (g) Is $\mu^- \rightarrow e^-e^-e^+\nu_\mu\bar{\nu}_e$ allowed?

(a)

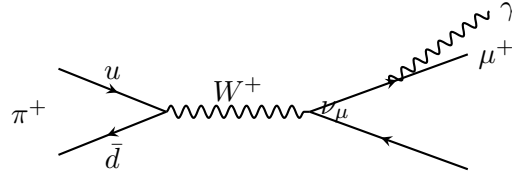
$$\Gamma = \frac{2\pi}{\hbar}|M|^2\rho.$$

- Γ : transition rate from $|i\rangle$ to continuum of $|f\rangle$ ($= 1/\tau$).
- $M = \langle f|H'|i\rangle$: matrix element of perturbing Hamiltonian.
- $\rho = dn/dE_f$: density of final states at the conserved energy.

Validity: $|M|^2t/\hbar \ll 1$ and ρ smooth (continuum).

(b)

$\pi^+ \rightarrow \mu^+\nu_\mu\gamma$: photon radiated off μ^+ (or pion / W):



Branching-ratio scaling. Two-body decay $\pi^+ \rightarrow \mu^+\nu_\mu$ is helicity-suppressed:

$$\Gamma_2 \propto G_F^2 f_\pi^2 m_\pi m_\mu^2 (1 - m_\mu^2/m_\pi^2)^2.$$

Photon emission removes the chirality flip; cost is α and three-body phase space.

Two-body phase space:

$$\Phi_2 \sim \frac{1}{8\pi}.$$

Three-body phase space (relativistic, masses small):

$$\Phi_3 \sim \frac{m_\pi^2}{192\pi^3}.$$

Form ratio of widths, removing m_μ^2 helicity factor and inserting α :

$$\frac{\Gamma(\pi^+ \rightarrow \mu^+\nu_\mu\gamma)}{\Gamma(\pi^+ \rightarrow \mu^+\nu_\mu)} \sim \alpha \cdot \frac{1}{m_\mu^2} \cdot \frac{\Phi_3}{\Phi_2}.$$

Substitute the phase-space estimates:

$$\frac{\Gamma_3}{\Gamma_2} \sim \frac{\alpha}{m_\mu^2} \cdot \frac{m_\pi^2/(192\pi^3)}{1/(8\pi)} = \frac{\alpha m_\pi^2}{24\pi^2 m_\mu^2}.$$

Include the $(1 - m_\mu^2/m_\pi^2)^{-2}$ from removing the helicity-flip kinematic factor:

$$\text{BR} \sim \frac{\alpha}{24\pi^2} \frac{m_\pi^2}{m_\mu^2} (1 - m_\mu^2/m_\pi^2)^{-2}.$$

Plug in $\alpha = 1/137 = 7.3 \times 10^{-3}$, $24\pi^2 \approx 237$, $m_\pi/m_\mu = 139.6/105.7 = 1.32$:

$$(m_\pi/m_\mu)^2 = 1.74.$$

Evaluate the helicity-flip removal factor with $(m_\mu/m_\pi)^2 = 0.573$:

$$(1 - 0.573)^{-2} = (0.427)^{-2} = 5.5.$$

Combine:

$$\text{BR} \sim \frac{7.3 \times 10^{-3}}{237} \cdot 1.74 \cdot 5.5 \approx 3 \times 10^{-4}.$$

Measured: 2×10^{-4} .

Assumptions: same $V - A$ Hamiltonian, universal QED coupling, helicity flip removed, slowly varying form factors.

(c)

Both purely leptonic weak: $\Gamma \propto G_F^2 m^5 / (192\pi^3)$.

Tau has multiple open channels (Cabibbo-suppressed $s\bar{u}$ neglected):

$$N_{\text{open}} = 1 (e\bar{\nu}_e\nu_\tau) + 1 (\mu\bar{\nu}_\mu\nu_\tau) + N_c (d\bar{u}\nu_\tau) \approx 1 + 1 + 3 = 5.$$

Form ratio:

$$\frac{\tau_\tau}{\tau_\mu} = \frac{1}{N_{\text{open}}} \left(\frac{m_\mu}{m_\tau} \right)^5.$$

Numerical with $m_\mu = 105.7$, $m_\tau = 1777$ MeV:

$$m_\mu/m_\tau = 0.0595.$$

Raise to 5th power:

$$(m_\mu/m_\tau)^5 = 7.4 \times 10^{-7}.$$

Multiply by τ_μ/N_{open} :

$$\tau_\tau = (2.2 \times 10^{-6} \text{ s}) \times \frac{7.4 \times 10^{-7}}{5}.$$

Evaluate:

$$\tau_\tau \approx 3.3 \times 10^{-13} \text{ s}.$$

(Measured: 2.9×10^{-13} s; QCD enhances $N_{\text{open}} \rightarrow \sim 5.6$.)

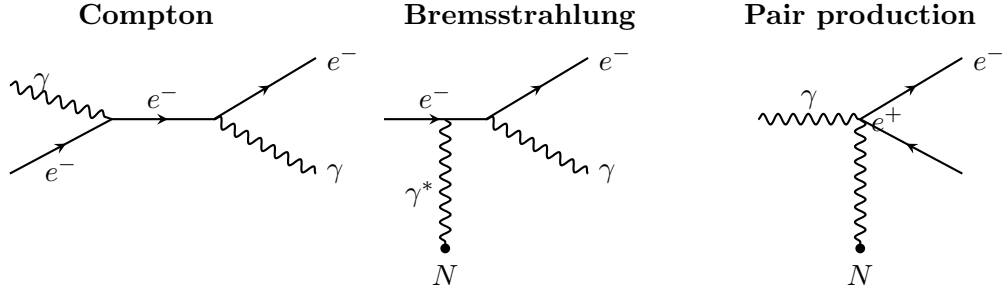
(d)

M in FGR is the sum of Feynman amplitudes. Components:

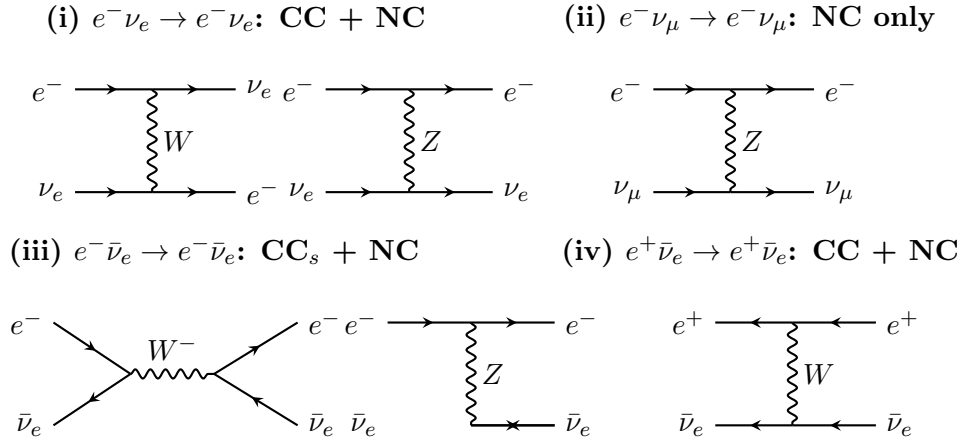
- **External lines:** physical incoming/outgoing particles, on-shell.
- **Propagators (internal lines):** virtual quanta, factor $i/(p^2 - m^2 + i\epsilon)$.
- **Vertices:** coupling $\times$ Dirac structure (e.g. $-ie\gamma^\mu$ for QED); enforce 4-momentum conservation.
- **Loops:** $\int d^4k/(2\pi)^4$.

Square, average over initial spins, sum over final spins, fold in $\rho \Rightarrow \Gamma$.

(e)



(f)



All weak: $\sigma \propto G_F^2 s$ at $\sqrt{s} \ll M_W$. Channels available:

- (i) CC (t -channel W) + NC (Z), constructive interference $\Rightarrow$ largest.
- (iv) CP image of (i) $\Rightarrow$ equal to (i) by CP invariance.
- (iii) CC (s -channel W) + NC, smaller interference than (i).
- (ii) NC only, no CC contribution $\Rightarrow$ smallest.

$$\sigma_{(i)} \approx \sigma_{(iv)} > \sigma_{(iii)} > \sigma_{(ii)}.$$

(g)

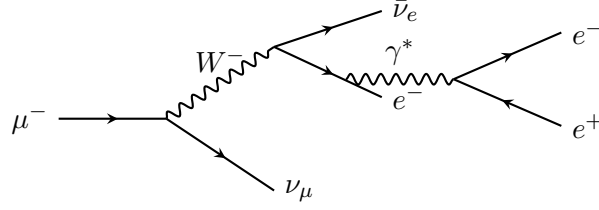
Process $\mu^- \rightarrow e^- e^- e^+ \nu_\mu \bar{\nu}_e$. Quantum numbers:

$$Q : -1 \rightarrow -1 - 1 + 1 + 0 + 0 = -1. \checkmark$$

$$L_\mu : +1 \rightarrow +1. \checkmark$$

$$L_e : 0 \rightarrow (+1) + (+1) + (-1) + (-1) = 0. \checkmark$$

Allowed. It is ordinary muon decay with an extra internal $\gamma^* \rightarrow e^+ e^-$:



BR suppressed by $\alpha^2 \sim 5 \times 10^{-5}$ relative to ordinary muon decay.

Question 3 — β , α decay; chain kinetics

Problem. (a) Q -values for $\beta^\mp$ and EC in atomic masses. (b) $T_\alpha(Q, A)$. (c) Assumptions in Gamow factor. (d) For $a \rightarrow b \rightarrow c$: (i) α, β , (ii) $t_{\max}$ when $N_{b0} = 0$, (iii) $N_c(t)$.

(a)

Atomic masses $M(P), M(D)$; nuclear $M_{\text{nuc}}(P) = M(P) - Zm_e$; (anti) ν massless.

β^- : ${}^A_Z P \rightarrow {}^A_{Z+1} D + e^- + \bar{\nu}_e$. Nuclear-level energy balance:

$$Q = [M_{\text{nuc}}(P) - M_{\text{nuc}}(D) - m_e]c^2.$$

Replace $M_{\text{nuc}} \rightarrow M - Zm_e$ (D has $Z + 1$ electrons):

$$Q = [M(P) - Zm_e - M(D) + (Z + 1)m_e - m_e]c^2.$$

Cancel electron terms:

$$Q_{\beta^-} = [M(P) - M(D)]c^2.$$

β^+ : ${}^A_Z P \rightarrow {}^A_{Z-1} D + e^+ + \nu_e$. Nuclear-level:

$$Q = [M_{\text{nuc}}(P) - M_{\text{nuc}}(D) - m_e]c^2.$$

Replace M_{nuc} atomic (D has $Z - 1$ electrons):

$$Q = [M(P) - Zm_e - M(D) + (Z - 1)m_e - m_e]c^2.$$

Collect electron terms:

$$Q_{\beta^+} = [M(P) - M(D) - 2m_e]c^2.$$

Electron capture: ${}^A_Z P + e^-_{\text{bound}} \rightarrow {}^A_{Z-1} D + \nu_e$. Bound K -shell electron has energy $m_e c^2 - B_e$. Nuclear-level:

$$Q = [M_{\text{nuc}}(P) + m_e - M_{\text{nuc}}(D)]c^2 - B_e.$$

Replace by atomic masses:

$$Q = [M(P) - Zm_e + m_e - M(D) + (Z - 1)m_e]c^2 - B_e.$$

Cancel electron terms:

$$Q_{\text{EC}} = [M(P) - M(D)]c^2 - B_e.$$

EC remains open when β^+ closes (no $2m_e$ penalty, only $B_e \sim \text{keV}$).

(b)

${}^A_ZP \rightarrow {}^{A-4}_{Z-2}D + \alpha$ from P at rest. Momentum conservation:

$$|\vec{p}_\alpha| = |\vec{p}_D| \equiv p.$$

Non-relativistic ($Q \ll M_D$):

$$T_\alpha = \frac{p^2}{2m_\alpha}, \quad T_D = \frac{p^2}{2M_D}.$$

Form ratio:

$$\frac{T_D}{T_\alpha} = \frac{m_\alpha}{M_D}.$$

Use $Q = T_\alpha + T_D = T_\alpha(1 + m_\alpha/M_D)$:

$$T_\alpha = Q \cdot \frac{M_D}{m_\alpha + M_D}.$$

Use $m_\alpha \approx 4u$, $M_D \approx (A - 4)u$:

$$T_\alpha = Q \cdot \frac{A - 4}{A}.$$

Rewrite:

$$\boxed{T_\alpha = Q \left(1 - \frac{4}{A}\right)}.$$

(c)

$\tau \propto \exp[Ze^2/(\epsilon_0 \hbar v)]$ from WKB tunnelling of pre-formed α through Coulomb barrier. Assumptions:

1. Pre-formed α cluster inside parent, assault frequency $\nu_0 \sim v/(2R)$.
2. Daughter as point charge $(Z - 2)e$; Coulomb barrier outside R only.
3. Centrifugal barrier neglected (s-wave).
4. WKB: $V - E$ slow on de Broglie scale.
5. Low-velocity limit: inner turning point $\rightarrow 0$; leading exponent $Ze^2/(\epsilon_0 \hbar v)$.
6. Energy-independent prefactor; all (Z, v) -dependence in exponent.

Result: Geiger–Nuttall straight line over ~ 24 decades.

(d)(i)

Bateman equations:

$$\frac{dN_a}{dt} = -\lambda_a N_a, \quad \frac{dN_b}{dt} = \lambda_a N_a - \lambda_b N_b.$$

Integrate first equation:

$$N_a(t) = N_{a0} e^{-\lambda_a t}.$$

Substitute into second:

$$\frac{dN_b}{dt} + \lambda_b N_b = \lambda_a N_{a0} e^{-\lambda_a t}.$$

Multiply by integrating factor $e^{\lambda_b t}$:

$$\frac{d}{dt}(N_b e^{\lambda_b t}) = \lambda_a N_{a0} e^{(\lambda_b - \lambda_a)t}.$$

Integrate:

$$N_b e^{\lambda_b t} = \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a} e^{(\lambda_b - \lambda_a)t} + K.$$

Divide by $e^{\lambda_b t}$:

$$N_b(t) = \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a} e^{-\lambda_a t} + K e^{-\lambda_b t}.$$

Apply $N_b(0) = N_{b0}$:

$$N_{b0} = \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a} + K.$$

Solve for K :

$$K = N_{b0} - \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a}.$$

Read off coefficients of $e^{-\lambda_a t}, e^{-\lambda_b t}$:

$$\boxed{\alpha = \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a}, \quad \beta = N_{b0} - \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a}.$$

(d)(ii)

Set $N_{b0} = 0$ so $K = -\lambda_a N_{a0}/(\lambda_b - \lambda_a)$:

$$N_b(t) = \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a} (e^{-\lambda_a t} - e^{-\lambda_b t}).$$

Differentiate with respect to t :

$$\frac{dN_b}{dt} = \frac{\lambda_a N_{a0}}{\lambda_b - \lambda_a} (-\lambda_a e^{-\lambda_a t} + \lambda_b e^{-\lambda_b t}).$$

Set the bracket to zero at $t = t_{\max}$:

$$\lambda_b e^{-\lambda_b t_{\max}} = \lambda_a e^{-\lambda_a t_{\max}}.$$

Divide both sides:

$$e^{(\lambda_b - \lambda_a)t_{\max}} = \frac{\lambda_b}{\lambda_a}.$$

Take logarithm:

$$\boxed{t_{\max} = \frac{\ln(\lambda_b/\lambda_a)}{\lambda_b - \lambda_a}.$$

(d)(iii)

c stable: $dN_c/dt = \lambda_b N_b$. Substitute N_b from (d)(ii):

$$\frac{dN_c}{dt} = \frac{\lambda_a \lambda_b N_{a0}}{\lambda_b - \lambda_a} (e^{-\lambda_a t} - e^{-\lambda_b t}).$$

Integrate term-by-term from 0 to t with $N_c(0) = 0$:

$$\int_0^t e^{-\lambda_a t'} dt' = \frac{1 - e^{-\lambda_a t}}{\lambda_a}, \quad \int_0^t e^{-\lambda_b t'} dt' = \frac{1 - e^{-\lambda_b t}}{\lambda_b}.$$

Combine:

$$N_c(t) = \frac{\lambda_a \lambda_b N_{a0}}{\lambda_b - \lambda_a} \left[\frac{1 - e^{-\lambda_a t}}{\lambda_a} - \frac{1 - e^{-\lambda_b t}}{\lambda_b} \right].$$

Combine over common denominator:

$$N_c(t) = \frac{N_{a0}}{\lambda_b - \lambda_a} [\lambda_b(1 - e^{-\lambda_a t}) - \lambda_a(1 - e^{-\lambda_b t})].$$

Expand and group the 1's:

$$N_c(t) = \frac{N_{a0}}{\lambda_b - \lambda_a} [(\lambda_b - \lambda_a) - (\lambda_b e^{-\lambda_a t} - \lambda_a e^{-\lambda_b t})].$$

Factor N_{a0} out:

$$N_c(t) = N_{a0} \left[1 - \frac{\lambda_b e^{-\lambda_a t} - \lambda_a e^{-\lambda_b t}}{\lambda_b - \lambda_a} \right].$$

Check: $N_c(0) = 0$, $N_c(\infty) = N_{a0}$, $N_a + N_b + N_c = N_{a0}$.

Question 4 — Eightfold way: $J = 1/2$ octet, $J = 0$ nonet

Problem. (a) Meaning of B, S, I_z ; reproduce octet pattern and identify X', Y' . (b) One state per edge, none at corners. (c) Two centre states. (d) Evidence for qqq . (e) $Q(I_z, S + B)$ relation; quark charges. (f) Σ^0 decay. (g) $J = 0$ meson nonet.

(a)

Definitions:

- $B = (n_q - n_{\bar{q}})/3$: each quark $+1/3$, antiquark $-1/3$.
- $S = -(n_s - n_{\bar{s}})$: s has $S = -1$ (sign convention).
- I_z : SU(2) flavour third component; $u : +1/2$, $d : -1/2$, $s : 0$.

For qqq from $\{u, d, s\}$: $S + B = 1 - n_s$. Three rows: $n_s = 0, 1, 2$. Within row, $I_z = (n_u - n_d)/2$.

State	Quark content	$(I_z, S + B)$
p	uud	$(+1/2, +1)$
n	udd	$(-1/2, +1)$
Σ^+	uus	$(+1, 0)$
Σ^0	uds	$(0, 0)$
Λ^0	uds	$(0, 0)$
Σ^-	dds	$(-1, 0)$
Ξ^0	uss	$(+1/2, -1)$
Ξ^-	dss	$(-1/2, -1)$

Identify:

$$X' = n = udd, \quad Y' = p = uud.$$

(b)

Edge positions: two flavours among three, e.g. $(I_z, S+B) = (+1, 0)$ requires uus . Total wavefunction antisymmetric; colour singlet antisymmetric $\Rightarrow$ spin $\otimes$ flavour $\otimes$ space symmetric. With S -wave (spatial symmetric), spin $\otimes$ flavour symmetric. For mixed-symmetry flavour (uus) and $J = 1/2$, exactly one symmetric combination exists $\Rightarrow$ **one state**.

Corners: uuu, ddd, sss have totally symmetric flavour, requiring totally symmetric spin $\Rightarrow J = 3/2$ only (decuplet $\Delta^{++}, \Delta^-, \Omega^-$). **No $J = 1/2$ state.**

(c)

Centre $(0, 0)$: flavour uds , three distinct flavours admit two independent mixed-symmetry combinations under three-body permutation:

- $I_{ud} = 1$ (symmetric in ud): Σ^0 , $m = 1192.6$ MeV.
- $I_{ud} = 0$ (antisymmetric in ud): Λ^0 , $m = 1115.7$ MeV.

Different exchange interactions $\Rightarrow$ different masses.

(d)

Evidence for qqq :

- $B = +1$ conserved (pair production with antibaryon); excludes $q\bar{q}$.
- Half-integer spin $J = 1/2$: requires odd fermion count.
- Magnetic moments: SU(6) prediction $\mu_n/\mu_p = -2/3$ confirmed.
- Deep inelastic scattering: three valence quarks with charges $+2/3, -1/3, -1/3$.

(e)

Inspect table: Q matches $I_z + (S+B)/2$ for every entry.

$$Q = I_z + \frac{1}{2}(S+B) \quad (\text{Gell-Mann-Nishijima}).$$

Apply to quarks ($B = 1/3$ each, $S = -1$ for s):

$$Q_u = +\frac{1}{2} + \frac{1}{2}\left(0 + \frac{1}{3}\right) = +\frac{2}{3}.$$

$$Q_d = -\frac{1}{2} + \frac{1}{2}\left(0 + \frac{1}{3}\right) = -\frac{1}{3}.$$

$$Q_s = 0 + \frac{1}{2}\left(-1 + \frac{1}{3}\right) = -\frac{1}{3}.$$

$$Q_u = +\frac{2}{3}, \quad Q_d = -\frac{1}{3}, \quad Q_s = -\frac{1}{3}.$$

(f)

Σ^0 and Λ^0 both uds , $J^P = \frac{1}{2}^+$: no flavour change. $\Delta m = 76.9 \text{ MeV} < m_{\pi^0}$ so no on-shell hadronic mode. Strong decay forbidden; EM dominates:

$$\boxed{\Sigma^0 \rightarrow \Lambda^0 + \gamma \text{ (M1, } \tau \sim 10^{-19} \text{ s).}}$$

$\Sigma^0 \rightarrow n\pi^0, p\pi^-$ are weak (flavour-changing, $\Delta S = 1$) and kinematically open, but the much faster EM mode ($\sim 10^{10}$ times faster) dominates the lifetime entirely.

Photon energy. Energy/momentum conservation in Σ^0 rest frame:

$$E_\gamma + E_\Lambda = m_{\Sigma^0}, \quad |\vec{p}_\gamma| = |\vec{p}_\Lambda| = E_\gamma.$$

Square $E_\Lambda = m_{\Sigma^0} - E_\gamma$ and use $E_\Lambda^2 = m_\Lambda^2 + E_\gamma^2$:

$$m_\Lambda^2 + E_\gamma^2 = (m_{\Sigma^0} - E_\gamma)^2.$$

Expand right side:

$$m_\Lambda^2 + E_\gamma^2 = m_{\Sigma^0}^2 - 2m_{\Sigma^0}E_\gamma + E_\gamma^2.$$

Cancel E_γ^2 :

$$m_\Lambda^2 = m_{\Sigma^0}^2 - 2m_{\Sigma^0}E_\gamma.$$

Solve for E_γ :

$$\boxed{E_\gamma = \frac{m_{\Sigma^0}^2 - m_\Lambda^2}{2m_{\Sigma^0}}c^2.}$$

Compute masses squared:

$$m_{\Sigma^0}^2 = 1192.6^2 = 1.4223 \times 10^6 \text{ MeV}^2.$$

$$m_\Lambda^2 = 1115.7^2 = 1.2448 \times 10^6 \text{ MeV}^2.$$

Subtract:

$$m_{\Sigma^0}^2 - m_\Lambda^2 = 1.775 \times 10^5 \text{ MeV}^2.$$

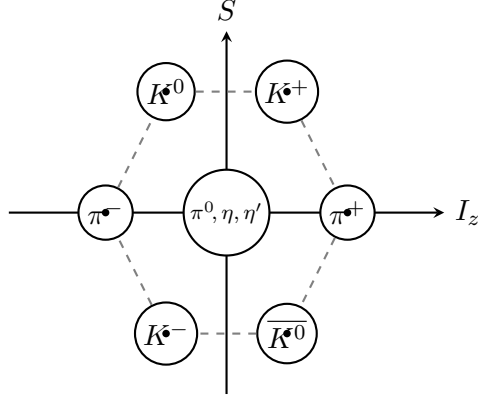
Divide by $2m_{\Sigma^0}$:

$$E_\gamma = \frac{1.775 \times 10^5}{2385.2} \text{ MeV} \approx 74.4 \text{ MeV}.$$

(g)

Mesons: $q\bar{q}$, $3 \otimes \bar{3} = 8 \oplus 1$. For $J = 0$ ground-state pseudoscalars:

State	Content	(I_z, S)
K^+	$u\bar{s}$	$(+1/2, +1)$
K^0	$d\bar{s}$	$(-1/2, +1)$
π^+	$u\bar{d}$	$(+1, 0)$
π^0	$\frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d})$	$(0, 0)$
η	$\approx \frac{1}{\sqrt{6}}(u\bar{u} + d\bar{d} - 2s\bar{s})$	$(0, 0)$
η'	$\approx \frac{1}{\sqrt{3}}(u\bar{u} + d\bar{d} + s\bar{s})$	$(0, 0)$
π^-	$d\bar{u}$	$(-1, 0)$
$\bar{K}^0$	$s\bar{d}$	$(+1/2, -1)$
K^-	$s\bar{u}$	$(-1/2, -1)$



Pions: $\pi^\pm, \pi^0$. Kaons: $K^\pm, K^0, \bar{K}^0$.

Three centre states ($I_z = 0, S = 0$), in SU(3) limit:

- $\pi^0 = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d})$: $I = 1$ octet member.
- $\eta_8 = \frac{1}{\sqrt{6}}(u\bar{u} + d\bar{d} - 2s\bar{s})$: $I = 0$ octet member.
- $\eta_1 = \frac{1}{\sqrt{3}}(u\bar{u} + d\bar{d} + s\bar{s})$: SU(3) singlet.

SU(3) breaking ($m_s > m_{u,d}$) mixes η_8, η_1 into physical η (548), η' (958) MeV. π^0 is unmixed (different I). Anomalously heavy η' from $U(1)_A$ axial anomaly.